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q_k^i : # jobs at server i at time k

A_k : Indicator if a job arrive at time k

d_k^i : departure from server i at time k

a_k : server to which a job is routed.

$$q_{k+1}^i = \left[q_k^i + A_k \mathbb{I}_{a_k=i} - d_k^i \right]^+$$

$$s_k = \begin{pmatrix} q_k^1 \\ q_k^2 \\ q_k^3 \end{pmatrix}$$

$$r(s,a) = q_k^1 + q_k^2 + q_k^3$$

Large Language Models:

x : prompt (expressed in tokens)

$y_1, y_2, \ldots, y_N$ (output tokens)

θ : parameters of an LLM

LLM: $p_\theta(y_1|x)\, p_\theta(y_2|y_1,x) \cdots p_\theta(y_N|y_1,\ldots,y_{N-1},x)$

$s_0 = x$

$s_1 = (x, y_1)$

$\vdots$

$s_N = (x, y_1, \ldots, y_{N-1})$

$a = \theta$

r : is difficult to explain now, we will revisit this example later.

Assumptions in this course:

- s_k, a_k take values in a discrete, finite set
- In this case, we can describe the system by a controlled Markov chain a.k.a. Markov Decision process:

$$P_{ij}(a) \text{ or } P(j|i,a) = P\left(s_{k+1}=j \,\middle|\, s_k=i, a_k=a\right)$$

Goal is to choose a policy $a_k = \pi(s_k)$, function of the state to maximize one of several possible objectives:

- $E\left(\sum_{k=0}^{T-1} r(s_k,a_k)\right)$: finite horizon problem with horizon = T
- $E\left(\sum_{k=0}^{\tau-1} r(s_k,a_k)\right)$: τ: random time when an goal is reached (e.g., Rubik's cube is solved)
- $E\left(\sum_{k=0}^{\infty} r(s_k,a_k)\right)$: infinite-horizon total reward
- $E\left(\sum_{k=0}^{\infty} \alpha^k r(s_k,a_k)\right)$: discounted reward $0 < \alpha < 1$
- $\lim_{T\to\infty} \frac{1}{T} E\left(\sum_{k=0}^{T-1} r(s_k,a_k)\right)$: average reward, infinite horizon

- In this course, we will mainly focus on the discounted reward problem.
- stochastic control | MDP theory : finding the optimal control policy when f is known
- Reinforcement learning (RL) : finding the optimal policy when f is unknown.